

28.4 Support recovery

Good prediction does not require the right variables: the bounds of [Section 28.3](#) allow false variables with small coefficients and missed true ones. Often, though, the variables are the object of interest. This section asks when the lasso selects *exactly* the right variables with the right signs. The answer needs a condition on the design much stronger than those of [Section 28.2](#), and it fails in simple, realistic examples.

28.4.1 Uniqueness

The question “which variables does the lasso select?” needs an answer that does not depend on which minimizer is reported.

Lemma 28.4.1. Uniqueness of the lasso fit

- (a) All lasso estimates at a given λ have the same fitted values $\mathbf{X}\hat{\boldsymbol{\beta}}_\lambda$ and the same ℓ_1 norm. Hence they share the vector $\hat{\mathbf{z}} = \mathbf{X}^\top(\mathbf{Y} - \mathbf{X}\hat{\boldsymbol{\beta}}_\lambda)/(n\lambda)$ of Eq. 28.3.2.
- (b) Let $\hat{\boldsymbol{\beta}}_\lambda$ be a lasso estimate with support $\hat{S}$. If $|\hat{z}_j| < 1$ for every $j \notin \hat{S}$ and $\mathbf{X}_{\hat{S}}$ has full column rank, then $\hat{\boldsymbol{\beta}}_\lambda$ is the only lasso estimate.

Proof.

- (a) The first statement is Theorem 27.4.1(c). The vector $\hat{\mathbf{z}}$ depends on $\hat{\boldsymbol{\beta}}_\lambda$ only through $\mathbf{X}\hat{\boldsymbol{\beta}}_\lambda$.
- (b) Let $\mathbf{b}$ be any lasso estimate. By (a) its optimality conditions Eq. 28.3.2 hold with the same $\hat{\mathbf{z}}$. If $b_j \neq 0$ then $\hat{z}_j = \text{sign}(b_j)$, so $|\hat{z}_j| = 1$ and $j \in \hat{S}$. Thus $\mathbf{b}$ is supported on $\hat{S}$, and $\mathbf{X}_{\hat{S}}\mathbf{b}_{\hat{S}} = \mathbf{X}\mathbf{b} = \mathbf{X}\hat{\boldsymbol{\beta}}_\lambda = \mathbf{X}_{\hat{S}}\hat{\boldsymbol{\beta}}_{\lambda, \hat{S}}$. Full column rank of $\mathbf{X}_{\hat{S}}$ gives $\mathbf{b} = \hat{\boldsymbol{\beta}}_\lambda$. ■

28.4.2 The irrepresentable condition

When will the lasso choose exactly S ? Suppose for a moment that it does, with the correct signs $\mathbf{z}_S = \text{sign}(\boldsymbol{\beta}_S)$. Then Eq. 28.3.2 on S determines the estimate: $\mathbf{X}_S^\top(\mathbf{Y} - \mathbf{X}_S\hat{\boldsymbol{\beta}}_S)/n = \lambda\mathbf{z}_S$, so

$$\begin{aligned}\hat{\boldsymbol{\beta}}_S &= (\mathbf{X}_S^\top \mathbf{X}_S)^{-1}(\mathbf{X}_S^\top \mathbf{Y} - n\lambda\mathbf{z}_S) \\ &= \boldsymbol{\beta}_S + (\mathbf{X}_S^\top \mathbf{X}_S)^{-1} \mathbf{X}_S^\top \boldsymbol{\varepsilon} - \lambda \hat{\boldsymbol{\Sigma}}_{SS}^{-1} \mathbf{z}_S,\end{aligned}\tag{28.4.1}$$

where $\hat{\boldsymbol{\Sigma}}_{SS} = \mathbf{X}_S^\top \mathbf{X}_S/n$. The estimate is least squares on the true support, shifted towards zero by $\lambda \hat{\boldsymbol{\Sigma}}_{SS}^{-1} \mathbf{z}_S$. For each $k \notin S$, the optimality conditions also require $|\mathbf{x}_k^\top(\mathbf{Y} - \mathbf{X}_S\hat{\boldsymbol{\beta}}_S)| \leq n\lambda$. Since $\mathbf{Y} - \mathbf{X}_S\hat{\boldsymbol{\beta}}_S = (\mathbf{I} - \mathbf{M}_S)\boldsymbol{\varepsilon} + n\lambda\mathbf{X}_S(\mathbf{X}_S^\top \mathbf{X}_S)^{-1}\mathbf{z}_S$, where $\mathbf{M}_S$ projects onto $\mathcal{C}(\mathbf{X}_S)$, this reads

$$\left| \underbrace{\mathbf{x}_k^\top \mathbf{X}_S (\mathbf{X}_S^\top \mathbf{X}_S)^{-1} \mathbf{z}_S}_{a_k} + \frac{\mathbf{x}_k^\top (\mathbf{I} - \mathbf{M}_S) \boldsymbol{\varepsilon}}{n\lambda} \right| \leq 1.\tag{28.4.2}$$

The first term does not involve the noise at all. The vector $(\mathbf{X}_S^\top \mathbf{X}_S)^{-1} \mathbf{X}_S^\top \mathbf{x}_k$ holds the least squares coefficients of $\mathbf{x}_k$ on the active columns, and a_k is their sum weighted by the true signs.

If an inactive column is well explained by the active ones, with coefficients that line up with the signs of $\boldsymbol{\beta}_S$, then $|a_k|$ is large and the condition is hard to meet.

Definition 28.4.1. Irrepresentable and beta-min conditions

Let $\mathbf{X}_S$ have full column rank. The design satisfies the **irrepresentable condition** for $\boldsymbol{\beta}$ with margin $\eta \in (0, 1]$ if

$$\max_{k \notin S} |\mathbf{x}_k^\top \mathbf{X}_S (\mathbf{X}_S^\top \mathbf{X}_S)^{-1} \text{sign}(\boldsymbol{\beta}_S)| \leq 1 - \eta.$$

A **beta-min condition** is a lower bound on $\beta_{\min} = \min_{j \in S} |\beta_j|$.

The condition depends on the design and on the signs of the true coefficients, but not on their sizes. It holds with $\eta = 1$ when the inactive columns are orthogonal to the active ones. It is much stronger than the compatibility condition. van de Geer and Bühlmann (2009) compare the two and show that a uniform version of the irrepresentable condition implies compatibility; the example below shows that the converse fails.

28.4.3 Sufficient conditions

Theorem 28.4.2. Sign recovery by the lasso

Suppose $\lambda_{\min}(\hat{\boldsymbol{\Sigma}}_{SS}) \geq C_{\min} > 0$ and the irrepresentable condition holds with margin η . Let $\lambda > 0$, put $g = \lambda \|\hat{\boldsymbol{\Sigma}}_{SS}^{-1} \text{sign}(\boldsymbol{\beta}_S)\|_\infty$, and suppose $\beta_{\min} > g$. Consider the events

$$\mathcal{A} = \left\{ \max_{k \notin S} \frac{|\mathbf{x}_k^\top (\mathbf{I} - \mathbf{M}_S) \boldsymbol{\varepsilon}|}{n} < \eta \lambda \right\},$$

$$\mathcal{B} = \left\{ \|(\mathbf{X}_S^\top \mathbf{X}_S)^{-1} \mathbf{X}_S^\top \boldsymbol{\varepsilon}\|_\infty < \beta_{\min} - g \right\}.$$

- (a) On $\mathcal{A} \cap \mathcal{B}$ the lasso estimate is unique and $\text{sign}(\hat{\boldsymbol{\beta}}_\lambda) = \text{sign}(\boldsymbol{\beta})$. In particular its support is exactly S .
- (b) The probabilities of the complements satisfy

$$\Pr(\mathcal{A}^c) \leq 2(p - s) \exp\left\{-\frac{n\eta^2 \lambda^2}{2\sigma^2}\right\}, \quad \Pr(\mathcal{B}^c) \leq 2s \exp\left\{-\frac{nC_{\min}(\beta_{\min} - g)^2}{2\sigma^2}\right\}.$$

- (c) Consequently, if

$$\lambda = \frac{\sigma}{\eta} \sqrt{\frac{2 \log\{4(p - s)/\delta\}}{n}} \quad \text{and} \quad \beta_{\min} \geq g + \sigma \sqrt{\frac{2 \log(4s/\delta)}{nC_{\min}}},$$

the lasso recovers the signs of $\boldsymbol{\beta}$ with probability at least $1 - \delta$.

Proof.

- (a) Construct a candidate and check that it is optimal. This is the *primal–dual witness* argument. Let $\mathbf{z}_S = \text{sign}(\boldsymbol{\beta}_S)$, let $\check{\boldsymbol{\beta}}$ be zero off S with $\check{\boldsymbol{\beta}}_S$ given by the right side of Eq. 28.4.1, and let $\check{\mathbf{z}} = \mathbf{X}^\top (\mathbf{Y} - \mathbf{X}\check{\boldsymbol{\beta}})/(n\lambda)$.
On S . By construction $\mathbf{X}_S^\top (\mathbf{Y} - \mathbf{X}_S \check{\boldsymbol{\beta}}_S) = n\lambda \mathbf{z}_S$, so $\check{\mathbf{z}}_S = \mathbf{z}_S$. By Eq. 28.4.1, for $j \in S$,

$|\check{\beta}_j - \beta_j| \leq \|(\mathbf{X}_S^\top \mathbf{X}_S)^{-1} \mathbf{X}_S^\top \boldsymbol{\varepsilon}\|_\infty + g < \beta_{\min} \leq |\beta_j|$ on $\mathcal{B}$. So $\check{\beta}_j$ is nonzero with the sign of β_j , and $\check{z}_j = \text{sign}(\check{\beta}_j)$.

Off S . Since $\mathbf{X}\boldsymbol{\beta} = \mathbf{X}_S \boldsymbol{\beta}_S \in \mathcal{C}(\mathbf{X}_S)$, $(\mathbf{I} - \mathbf{M}_S)\mathbf{Y} = (\mathbf{I} - \mathbf{M}_S)\boldsymbol{\varepsilon}$, and

$$\mathbf{Y} - \mathbf{X}_S \check{\boldsymbol{\beta}}_S = \mathbf{Y} - \mathbf{M}_S \mathbf{Y} + n\lambda \mathbf{X}_S (\mathbf{X}_S^\top \mathbf{X}_S)^{-1} \mathbf{z}_S = (\mathbf{I} - \mathbf{M}_S)\boldsymbol{\varepsilon} + n\lambda \mathbf{X}_S (\mathbf{X}_S^\top \mathbf{X}_S)^{-1} \mathbf{z}_S.$$

So for $k \notin S$, $\check{z}_k = a_k + \mathbf{x}_k^\top (\mathbf{I} - \mathbf{M}_S)\boldsymbol{\varepsilon}/(n\lambda)$, with $|a_k| \leq 1 - \eta$ by the irrerepresentable condition. On $\mathcal{A}$, $|\check{z}_k| < 1 - \eta + \eta = 1$.

So $\check{\boldsymbol{\beta}}$ and $\check{\mathbf{z}}$ satisfy Eq. 28.3.2, and $\check{\boldsymbol{\beta}}$ is a lasso estimate, because the conditions are sufficient (Theorem 27.4.1(b)). Its support is S , $\mathbf{X}_S$ has full column rank because $C_{\min} > 0$, and $|\check{z}_k| < 1$ off S . By Lemma 28.4.1(b) it is the only lasso estimate.

- (b) $\mathbf{x}_k^\top (\mathbf{I} - \mathbf{M}_S)\boldsymbol{\varepsilon}/n$ is normal with mean zero and variance $\sigma^2 \|(\mathbf{I} - \mathbf{M}_S)\mathbf{x}_k\|^2/n^2 \leq \sigma^2 \|\mathbf{x}_k\|^2/n^2 = \sigma^2/n$. The j th entry of $(\mathbf{X}_S^\top \mathbf{X}_S)^{-1} \mathbf{X}_S^\top \boldsymbol{\varepsilon}$ is normal with mean zero and variance $\sigma^2 [(\mathbf{X}_S^\top \mathbf{X}_S)^{-1}]_{jj} \leq \sigma^2/(nC_{\min})$, by Theorem 1.7.7 applied to $(\mathbf{X}_S^\top \mathbf{X}_S)^{-1}$, whose largest eigenvalue is $1/(nC_{\min})$ at most. Apply Lemma 28.3.2 with $t = \eta\lambda$ and $t = \beta_{\min} - g$. The events in the lemma use $>$, and the events here use $\geq$, which have the same probability for normal variables with positive variance (and probability zero for those with zero variance).
- (c) With the stated λ , the first bound in (b) equals $\delta/2$. With the stated $\beta_{\min}$, the second is at most $\delta/2$.

■

The three hypotheses ask that the active columns be well conditioned ($C_{\min}$), that the inactive ones not be representable by them (η), and that the coefficients stand out from noise and shrinkage ($\beta_{\min}$). Since $g \leq \lambda\sqrt{s}/C_{\min}$, the last asks in the worst case for every nonzero coefficient to exceed a multiple of $\sigma\sqrt{s \log p}/n$. The theorem in this form is due to Wainwright (2009).

28.4.4 The condition is also necessary

Proposition 28.4.3. Failure of the irrerepresentable condition

Let $\mathbf{X}_S$ have full column rank, and suppose that for some $k \notin S$

$$|a_k| = |\mathbf{x}_k^\top \mathbf{X}_S (\mathbf{X}_S^\top \mathbf{X}_S)^{-1} \text{sign}(\boldsymbol{\beta}_S)| > 1.$$

Then the probability that for some $\lambda > 0$ some lasso estimate has $\text{sign}(\hat{\boldsymbol{\beta}}_\lambda) = \text{sign}(\boldsymbol{\beta})$ is at most $\frac{1}{2}$.

Proof. Suppose a lasso estimate at some $\lambda > 0$ has the signs of $\boldsymbol{\beta}$. Its support is S , so by the calculation leading to Eq. 28.4.2, $|a_k + W_k/(n\lambda)| \leq 1$, where $W_k = \mathbf{x}_k^\top (\mathbf{I} - \mathbf{M}_S)\boldsymbol{\varepsilon}$. Since $|a_k| > 1$, this forces $\text{sign}(a_k) W_k/(n\lambda) \leq 1 - |a_k| < 0$, so $\text{sign}(a_k) W_k < 0$. The event in question is therefore contained in $\{\text{sign}(a_k) W_k < 0\}$, which does not depend on λ . W_k is normal with mean zero, so this event has probability $\frac{1}{2}$ if $(\mathbf{I} - \mathbf{M}_S)\mathbf{x}_k \neq \mathbf{0}$, and probability zero otherwise.

■

The argument is due to Zhao and Yu (2006), who showed in addition that the condition with ≤ 1 is necessary for sign consistency. The bound $\frac{1}{2}$ does not improve with more data. Sampling

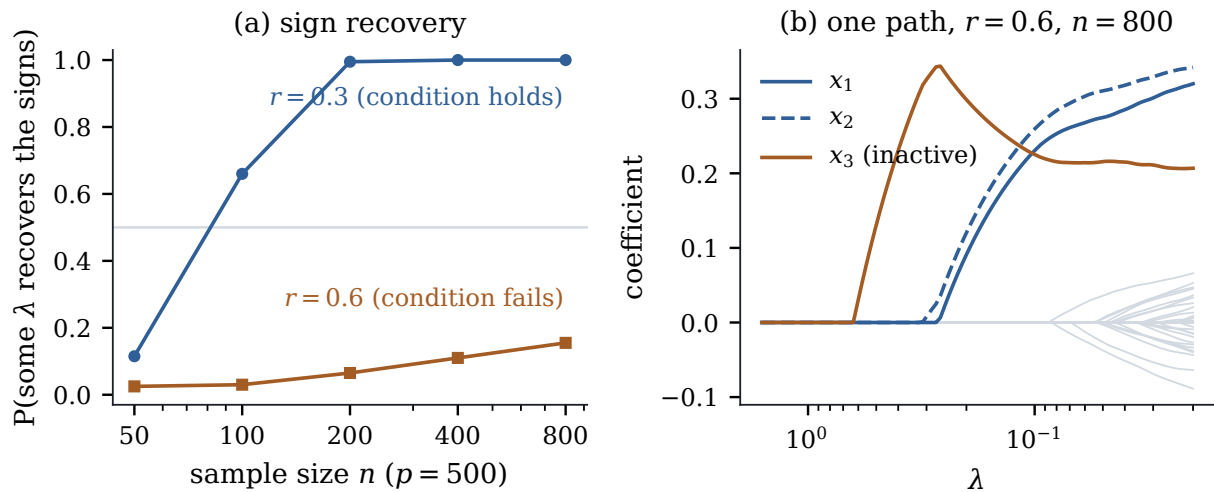


Figure 28.4.1: Support recovery with $p = 500$, two active regressors and an inactive regressor correlated r with both. (a) Fraction of data sets in which some penalty gives exactly the right signs, against n . (b) The lasso path of one data set with $r = 0.6$, $n = 800$: the inactive x_3 enters first and stays; grey lines are the noise variables.

more observations cannot repair a design in which an inactive variable imitates the active ones with coefficients aligned with their signs. In the noiseless limit the lasso *never* recovers the signs (Exercise (C1)).

Example. An inactive variable that looks active

Let x_1, x_2 be independent standard normal regressors, let x_3 have correlation r with each of them, and let $x_4, \dots, x_{500}$ be independent noise variables, with $Y = 0.5x_1 + 0.5x_2 + \varepsilon$ and $\sigma = 1$. The population coefficients of x_3 on (x_1, x_2) are (r, r) , so $a_3 = 2r$ (Exercise (A1)). With $r = 0.3$ the condition holds with margin 0.4; with the moderate correlation $r = 0.6$ it fails. Both population designs have $\lambda_{\min}(\Sigma) > 0$, so both are benign for prediction.

Figure 28.4.1(a) shows, for each sample size, the fraction of 200 data sets (fixed design, new errors) for which *some* penalty on a fine grid gives exactly the right signs. When the condition holds, this rises from 0.12 at $n = 50$ to 0.66 at $n = 100$ and 1.000 at $n = 800$. When it fails, it rises only slowly, from 0.03 at $n = 100$ to 0.155 at $n = 800$, and it can never exceed the bound $\frac{1}{2}$ of Proposition 28.4.3, however large n is. The sample irrerepresentable quantities at $n = 800$ are 0.66 and 1.21.

Panel (b) shows why. Along the lasso path for one data set with $r = 0.6$ and $n = 800$, the inactive x_3 is the first variable to enter, because its inner product $\mathbf{x}_3^\top \mathbf{X}\boldsymbol{\beta}/n$ with the signal (the mean of the $\mathbf{x}_3^\top \mathbf{Y}/n$ that decides which variable enters first) is 0.62 in this sample (0.6 in the population) and exceeds that of x_1 and x_2 (0.5 each). It never leaves. The lasso finds x_1 and x_2 later, but always together with x_3 .

```
import numpy as np

def soft(z, t):
    """Soft thresholding  $S(z, t) = \text{sign}(z) \max(|z| - t, 0)$ ."""
    return np.sign(z) * np.maximum(np.abs(z) - t, 0.0)

def lasso_cd(X, Y, lam, B=None, tol=1e-9, max_sweeps=2000):
    """Minimize  $\|y - Xb\|_2^2 / (2n) + \text{lam} \|b\|_1$  for every column  $y$  of  $Y$ , by
```

```

coordinate descent."""
n, p = X.shape
Y = Y.reshape(n, -1)
B = np.zeros((p, Y.shape[1])) if B is None else B.copy()
R = Y - X @ B # residuals, one column per
response
scale = np.sum(X ** 2, axis=0) / n
active = np.arange(p)
for sweep in range(max_sweeps):
    change = 0.0
    for j in active:
        z = X[:, j] @ R / n + scale[j] * B[j]
        new = soft(z, lam) / scale[j]
        d = new - B[j]
        if np.any(d != 0):
            R -= np.outer(X[:, j], d)
            B[j] = new
            change = max(change, np.max(np.abs(d)))
    if change < tol:
        if len(active) == p: # converged over all
coordinates
            break
        active = np.arange(p) # check every coordinate once
more
    else:
        active = np.flatnonzero(np.any(B != 0, axis=1)) if sweep % 5 else np.
arange(p)
return B

def make_design(rng, n, p, r):
    """x_1, x_2 independent; x_3 correlated r with each; x_4, ..., x_p independent
    noise."""
    Z = rng.normal(size=(n, p))
    Z[:, 2] = r * Z[:, 0] + r * Z[:, 1] + np.sqrt(1 - 2 * r ** 2) * Z[:, 2]
    return Z / np.sqrt(np.sum(Z ** 2, axis=0) / n) # ||x_j||^2 = n

def irrepresentable(X, S, signs):
    """max over j outside S of |x_j^T X_S (X_S^T X_S)^{-1} sign(beta_S)|."""
    A = X[:, S]
    w = A @ np.linalg.solve(A.T @ A, signs)
    out = np.delete(X.T @ w, S)
    return np.max(np.abs(out))

def recovers_signs(X, Y, beta, lams):
    """For each column of Y: does some penalty in lams give exactly sign(beta)?"""
    ok = np.zeros(Y.shape[1], dtype=bool)
    B = None
    for lam in lams: # from large to small, warm
starts
        B = lasso_cd(X, Y, lam, B)
        ok |= np.all(np.sign(B) == np.sign(beta)[:, None], axis=0)
    return ok

rng = np.random.default_rng(2840)
n, p, sigma = 100, 200, 1.0
beta = np.zeros(p)
beta[:2] = 0.5
S = np.array([0, 1])

```

```

lams = np.geomspace(1.0, 0.1, 30)
for r in [0.3, 0.6]:
    X = make_design(rng, n, p, r)
    Y = (X @ beta)[:, None] + sigma * rng.normal(size=(n, 100))
    print(f"r = {r}: irrerepresentable quantity {irrerepresentable(X, S, np.ones(2)):.2f}
          ,",
          f" sign recovery in {recovers_signs(X, Y, beta, lams).mean():.2f} of 100
          data sets")

```

[Run this code online](#)

28.4.5 Screening instead of selection

A weaker goal, often enough in practice, is **screening**: the selected set should *contain* S , with false variables left for a second stage to remove. It needs only the restricted eigenvalue and a beta-min condition.

Proposition 28.4.4. Variable screening

Suppose $\lambda \geq 2\lambda_0$, the event $\mathcal{T}$ of Eq. 28.3.5 occurs, $\kappa^2(S) > 0$, and $\beta_{\min} > 3\lambda\sqrt{s}/\kappa^2(S)$. Then every lasso estimate has $\hat{S} \supseteq S$.

Proof. If $j \in S$ and $\hat{\beta}_{\lambda,j} = 0$, then $|\hat{\beta}_{\lambda,j} - \beta_j| = |\beta_j| \geq \beta_{\min}$. By Theorem 28.3.4(b), $|\hat{\beta}_{\lambda,j} - \beta_j| \leq \|\hat{\beta}_\lambda - \beta\| \leq 3\lambda\sqrt{s}/\kappa^2(S) < \beta_{\min}$, a contradiction. ■

In [Example \(An inactive variable that looks active\)](#) with $r = 0.6$, screening succeeds where selection fails: x_1 and x_2 are found, together with x_3 . Second stages build on this. The **thresholded lasso** discards small coefficients and needs no irrerepresentable condition ([Exercise \(B2\)](#)). The **adaptive lasso** of Zou (2006) refits with penalty weights $1/|\hat{\beta}_j|$ from a first stage, and **stability selection** (Meinshausen and Bühlmann 2010) keeps the variables selected in most subsamples. All need beta-min conditions; none can find a coefficient below the noise level.

Warning. The lasso's selected set is a random set that depends on small features of the data, especially among correlated regressors. Reporting it as “the important variables” without an assessment of its stability invites overinterpretation. [Section 28.5](#) shows what goes wrong if the selected set is then treated as fixed.

28.4.6 Exercises

28.4.6.1 A. Check your understanding

Exercise. A1

In the population version of [Example \(An inactive variable that looks active\)](#), with $\Sigma_{SS} = \mathbf{I}_2$ and $\Sigma_{3S} = (r, r)$, compute $\Sigma_{3S}\Sigma_{SS}^{-1}\text{sign}(\beta_S)$ for $\beta_S = (0.5, 0.5)$ and for $\beta_S = (0.5, -0.5)$. For which signs does the irrerepresentable condition fail?

Solution. $\Sigma_{3S}\Sigma_{SS}^{-1} = (r, r)$. With signs $(1, 1)$ the quantity is $2r$, which exceeds 1 for $r > \frac{1}{2}$. With signs $(1, -1)$ it is 0, so the condition holds with $\eta = 1$. The trouble arises only when x_3 imitates the combination $x_1 + x_2$ that appears in the signal.

28.4.6.2 B. Practice

Exercise. B1

Let $\Sigma = (1 - \rho)\mathbf{I} + \rho\mathbf{1}\mathbf{1}^\top$ with $0 \leq \rho < 1$. Show that $\Sigma_{kS}\Sigma_{SS}^{-1}\mathbf{z}_S = \rho(\mathbf{1}^\top\mathbf{z}_S)/\{1 + (s - 1)\rho\}$ for $k \notin S$, and conclude that the population irrepresentable condition holds for every sign pattern, with margin at least $(1 - \rho)/\{1 + (s - 1)\rho\}$.

Exercise. B2

(*Thresholded lasso.*) Under the assumptions of Theorem 28.3.4(b), let $B = 3\lambda\sqrt{s}/\kappa^2(S)$ and suppose $\beta_{\min} > 2B$. Show that $\{j : |\hat{\beta}_{\lambda,j}| > B\} = S$, and that the signs of the retained coefficients are correct. No irrepresentable condition is needed.

Solution. Every coordinate satisfies $|\hat{\beta}_{\lambda,j} - \beta_j| \leq \|\hat{\beta}_\lambda - \beta\| \leq B$. For $j \notin S$, $|\hat{\beta}_{\lambda,j}| \leq B$, so j is discarded. For $j \in S$, $|\hat{\beta}_{\lambda,j}| \geq |\beta_j| - B > 2B - B = B$, so j is kept, and $\hat{\beta}_{\lambda,j}$ lies within $B < |\beta_j|$ of β_j , so it has the same sign. The price is the stronger beta-min condition $\beta_{\min} > 6\lambda\sqrt{s}/\kappa^2(S)$ and a threshold that depends on the unknown κ and s .

28.4.6.3 C. Going deeper

Exercise. C1

Let $\sigma = 0$, so $\mathbf{Y} = \mathbf{X}\beta$, and let $\mathbf{X}_S$ have full column rank. Show that if $\max_{k \notin S} |a_k| < 1$, then every λ with $0 < \lambda < \beta_{\min}/\|\hat{\Sigma}_{SS}^{-1}\text{sign}(\beta_S)\|_\infty$ gives a unique lasso estimate with the correct signs, and that if $|a_k| > 1$ for some $k \notin S$, then no $\lambda > 0$ does.

Solution. With $\varepsilon = \mathbf{0}$, the events $\mathcal{A}$ and $\mathcal{B}$ of Theorem 28.4.2 hold whenever $\eta > 0$ and $\beta_{\min} > g$, which is the stated range of λ , with $\eta = 1 - \max_k |a_k|$. Part (a) of the theorem applies, and its proof uses no probability. Conversely, if some lasso estimate had the correct signs, Eq. 28.4.2 with $\varepsilon = \mathbf{0}$ would require $|a_k| \leq 1$ for all $k \notin S$. So with perfect data, the irrepresentable condition (in its weak form) is necessary and, with strict inequality, sufficient.